MIAMI UNIVERSITY

SIGHT: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

WSIC26-250206-009

INTERIM PROGRESS REPORT

Dr. Fadel Megahed

Miami University

Version 1.0

07/17/2026

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WSIC GRANT PERIOD OF PERFORMANCE: July 1, 2025 – June 30, 2027

REPORTING PERIOD: July 1, 2025 – June 30, 2026

REPORT DATE: July 17, 2026

PROJECT TITLE: SIGHT: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

WSIC PROJECT #: WSIC26-250206-009 **PRINCIPAL INVESTIGATOR:** Fadel M. Megahed

REPORT COMPLETED BY: Megahed, Carvalho, Shan, Yousif, Mayyas **E-MAIL:** fmegahed@miamioh.edu

1. PROJECT SCHEDULE STATUS: AHEAD OF SCHEDULE

The project is ahead of schedule. As shown in Figure 1, all Phase 1–3 tasks are complete, and core Phase 4 development (Tasks 4.1–4.3) met or exceeded target due dates. At the project midpoint, 11 of 15 scheduled tasks are complete, with significant progress on the remaining four. Five of nine success metrics have been achieved.

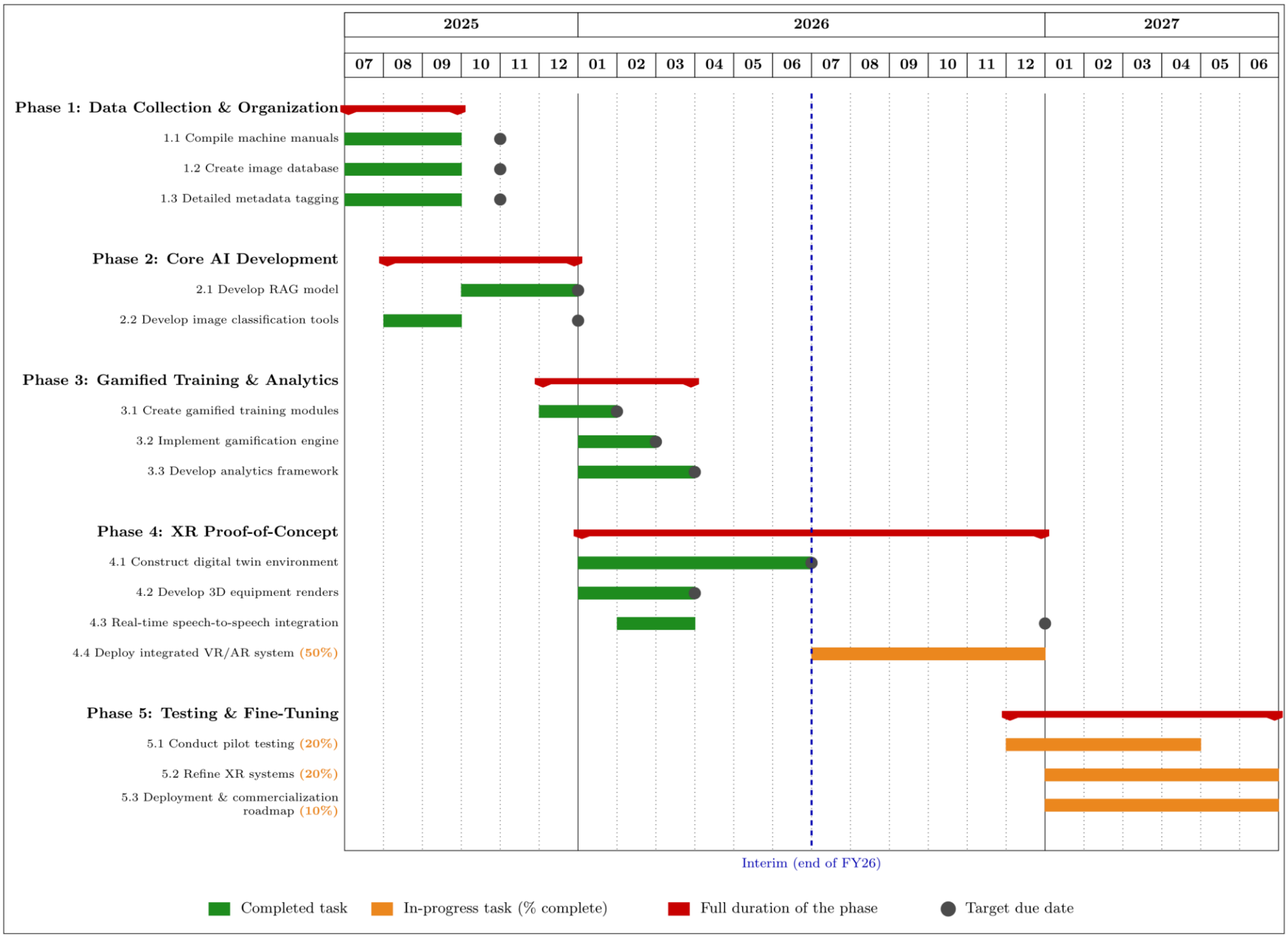


Figure 1. SIGHT project timeline and progress through the interim point (end of FY 26). Completed tasks are shown in green, in-progress tasks are shown in orange, and the phase's duration is reflected by the red anchored bars. The dark gray dot represents the task's scheduled completion date per our project management plan.

2. TOTAL PERCENTAGE OF PROJECT COMPLETION: >64.4%

The percentage completion is calculated by **averaging** our completed tasks, 73% (11 out of 15 tasks; see WBS Table on pp. 4–5 and the Gantt Chart on p. 2), and our completed success metrics, 55% (5 out of 9 metrics; see below).

Completed Success Metrics, Per Special Condition #1, WSIC-3 Section 2.5.5 (P.24 in "Exhibit D" of our contract)

Phase 1: Data Collection and Organization (Ends October 2025)

1. *Compile 100% standardized operational and safety documentation for the three selected target machines (Bridgeport Manual Mill, TL-1 CNC, and the Universal Cobot).*
2. *Create a comprehensive image database containing a minimum of 60 high-quality, metadata-tagged images per machine for the three target machines, covering multiple angles and operational scenarios.*

Phase 2: Core AI Development (Ends December 2025)

3. *Achieve at least 80% response accuracy for safety-critical information queries with the Retrieval-Augmented Generation (RAG) system, validated through expert review and user testing.*
4. *The AI image classification system will achieve at least 80% classification accuracy on a holdout dataset under standard lighting and at least 75% accuracy under varying environmental conditions.*

Phase 3: Gamified Training Script & Analytics Layer (Ends March 2026)

5. *Implement a functional analytics framework for tracking user interactions that successfully captures data from at least 90% of training interactions within 5 seconds per interaction.*

The remaining success metrics and their expected completion dates are provided in Section 7. As shown there, we have made considerable progress towards these metrics (consistent with what was shown in the Gantt Chart on p. 2).

3. PROJECT OVERVIEW

a. **Overview:** The SIGHT project aims to transform manufacturing safety training by integrating Artificial Intelligence (AI), Virtual Reality (VR), and Augmented Reality (AR). Its primary objective is to reduce preventable workplace injuries (particularly cuts, lacerations, and crush hazards) by delivering immersive, adaptive, and personalized training experiences. The project is creating a dynamic web-based VR platform for hazard recognition and an AR-based system that provides real-time, on-the-job safety coaching informed by IoT sensors and AI analytics.

b. **Scope:** The project spans five phases:

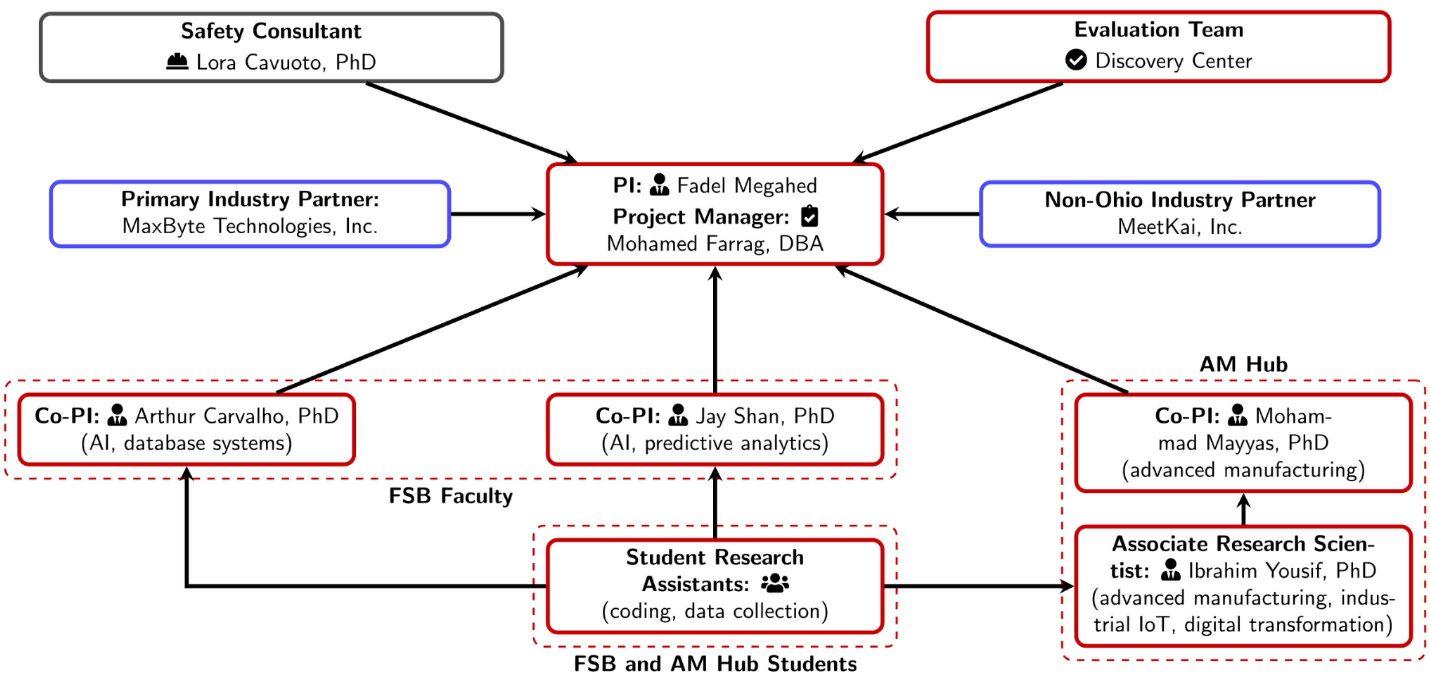
1. Data collection and organization to build AI-ready datasets.
2. Core AI development, including retrieval-augmented generation (RAG) and computer vision models.
3. Development of gamified training scripts and an analytics layer.
4. Design and implementation of XR proofs of concept (VR training modules and AR coaching overlays).
5. Testing, evaluation, and refinement with industry partners.

Expected outcomes include a publicly accessible VR training platform, validated AR safety modules deployed at Miami's Advanced Manufacturing Hub, measurable improvements in hazard recognition and retention, and a scalable roadmap for commercialization through partners MaxByte and MeetKai.

c. **Project Changes:** There have been three major changes since the original proposal.

1. The project has replaced Cincinnati Radiator (a non-funded industry partner) with Engineered Profiles, a leading Ohio-based supplier of extruded products. This change was made following Cincinnati Radiator's merger, which disrupted previous contacts. Engineered Profiles, headquartered in Columbus, will provide real-world feedback on the SIGHT platform and assist with participant recruitment. The replacement was formally completed within the WSIC's 10-day requirement, with no impact on budget or scope.
2. Due to increased teaching commitments, Dr. Reza Abrishambaf is no longer part of the project. Accordingly, his Senior Personnel role has been eliminated.
3. Dr. Ibrahim Yousif's position has been changed from Postdoctoral Researcher to Associate Research Scientist to reflect his expanded responsibilities following the departure of Dr. Abrishambaf.

d. **Organizational Chart**: The chart below reflects the above personnel changes:



Note: FSB refers to the Farmer School of Business, and AM stands for Advanced Manufacturing.

4. INTERIM ACTIVITY STATUS

The following table presents the current status of each aim and related tasks, based on our PMP.

Project Aims	Task No. & Description	Status	% Completed	Target Due Date	Date Completed

Aim 1 (VR)	1.1 Compile machine manuals	Complete	100%	10/31/2025	9/30/2025
	1.2 Create image database	Complete	100%	10/31/2025	9/5/2025
	1.3 Detailed metadata tagging	Complete	100%	10/31/2025	9/5/2025
Aim 1 (VR) Aim 2 (AR)	2.1 Develop RAG model	Complete	100%	12/31/2025	12/18/2025
	2.2 Develop image classification tools	Complete	100%	12/31/2025	9/12/2025
Aim 1 (VR)	3.1 Create gamified training modules	Complete	100%	01/31/2026	01/31/2026
Aim 2 (AR)	3.2 Implement gamification engine	Complete	100%	2/28/2026	02/28/2026
Aim 1 (VR)	3.3 Develop analytics framework	Complete	100%	03/31/2026	03/31/2026
Aim 2 (AR)	4.1 Construct a digital twin environment	Complete	100%	06/30/2026	06/30/2026
Aim 1 (VR) Aim 2 (AR)	4.2 Develop 3D equipment renders	Complete	100%	03/31/2026	03/17/2026
	4.3 Implement real-time speech-to-speech AI integration	Complete	100%	12/31/2026	03/17/2026
	4.4 Deploy Integrated VR and AR Training System	On Schedule	50%	12/31/2026	In progress
	5.1 Conduct pilot testing	On Schedule	20%	04/30/2027	In progress
	5.2 Refine XR systems	On Schedule	20%	06/30/2027	In progress
	5.3 Deployment & commercialization roadmap	On Schedule	10%	06/30/2027	In progress

5. PROJECT PROGRESS

A. **Methodologies:** The project advanced using our structured and phased methodology defined in both the *Project Proposal* and the Project Management plan. Our Miami U. SIGHT team is divided into two main sub-teams:

1. Advanced Manufacturing: Led by Dr. Mayyas (Co-PI), with core contributors from Dr. Yousif (Post Doc), Mr. Wise (GRA), and Mr. Lyublinski (student researcher).

2. Artificial Intelligence for Occupational Safety: Led by Dr. Carvalho (Co-PI), supported by Dr. Shan (Co-PI) and three student researchers (Hamilton, Singh, and White). Mr. Singh and Ms. White graduated in May 2026. We are recruiting new student researchers to join our team.

Dr. Megahed (PI) serves as the project owner, technical troubleshooter, and the link between both teams. Since mid-January 2026, Dr. Megahed has been working physically at the AM Hub 1-2 days per week to further optimize the alignment between the two sub-teams. Each sub-team has met weekly; the sub-team leads are often involved in individual meetings/calls to ensure that the student assistants are on track and to share information across teams. Dr. Farrag (PM) and the research team use Notion (Project Management Tool) to track our progress on the subtasks across different team members. Cross-team updates are performed in a more formal manner in our bi-weekly project meetings. We use a shared cloud environment (Google Drive) to retain copies and backups of each file.

Our team is supported by four industry partners, a safety consultant, and the Discovery Center for Evaluation, who play important roles in our progress. ~~An update on their engagement is provided in Bullet “C” in this section.~~

B. **Delays:** We have no current delays that impact our aims or milestones to report since we are ahead of schedule. However, we note the following updates on two items reported in Quarter 1 and 2:

1. IRB Revision: As noted in our Q2 report, we planned to submit an IRB revision to add (student) researchers to the approved protocol. Four team members (Dr. Ibrahim Yousif, Michael Wise, Austin Hamilton, and Leo Lyublinski) have successfully completed their CITI Training in preparation. Since our pilot testing with human participants (Task 5.1) is not scheduled to begin until December 2026, we have elected to defer the IRB amendment until our FY 2027 staffing plan is finalized, allowing us to consolidate all personnel changes into a single request to the IRB.
2. Invention Disclosure: In Q1 and Q2, we reported that the team was preparing an invention disclosure for our image-to-3D AI-driven rendering pipeline. Following additional testing and refinement, the invention disclosure was formally submitted to Miami University on April 10, 2026. In addition, a 2nd invention disclosure regarding the cognitive autonomous robotic assembly system was formally submitted to Miami University on June 8, 2026. It is currently moving through the process toward a provisional patent.

C. **IRB:** The project's IRB protocol (#02415r) was approved on November 20, 2025, as reported in Q2. As noted in Section 5B, four additional team members (Dr. Ibrahim Yousif, Michael Wise, Austin Hamilton, and Leo Lyublinski) completed their CITI Training; these are non-faculty team members who will participate in FY27. The formal IRB amendment to add these researchers will be submitted once our FY 2027 staffing plan is finalized, as pilot testing with human participants is not scheduled to begin until December 2026. We note that the 51-participant VR user testing conducted this quarter involved convenience-sampled, unmonitored web-based sessions. Individual response data was accessible only to the PI and MeetKai (as the platform developer), and only aggregated results were shared with the broader team.

6. TECHNICAL PROGRESS & CONSIDERATIONS

[Summarize technical updates during the reporting period, including:

- ***Technical Approach & Testing:*** *Describe technical approach, testing activities conducted, and outcomes. Report on any issues or barriers encountered. Provide updates on progress toward meeting identified testing and certification standards required for market readiness.*
- ***Ethical, Legal & Compliance Considerations:*** *Describe any ethical implications or legal issues that have emerged. This may include concerns related to intellectual property, data privacy, research ethics, or compliance with regulations.*

[Provide an updated testing/certification table to address all relevant testing and certification standards (e.g., OSHA standards, ADA, etc.) required for the innovation for commercialization. Identify the Testing/Certification Standard, Test Description, Test Status (e.g., planned, in progress, completed) and Test Result (e.g., Passed, To Be Determined, Failed or Other.]

Testing/Certification Standard	Test Description	Test Status	Test Result
[Example: Web Content Accessibility Guidelines (WCAG) 2.1 International standard for web content accessibility]	*[Performed a combination of automated, manual and functional testing by native screen reader users upon the agreed upon representative unique components and use cases of system]*	*[Planned]*	*[To be determined]*
<add more rows as needed>			

7. SUCCESS METRICS

[In the chart below, list the project success metrics (e.g., aims, milestones, etc.) and associated statuses. If applicable, describe any favorable developments that will enable project aims to be realized sooner. Provide the Success Metric and Description, Status (on schedule, ahead of schedule, behind schedule, complete or cancelled), Percent of Work Completed, and Projected Completion Date.]

Success Metric and Description	Status	% of Work Completed	Projected Completion Date
[Example: Create a product website that allows product accessibility for subscription service]	*[On schedule]*	*[50%]*	*[11/12/2026]*
<Add rows as needed>			

8. ADMINISTRATIVE STATUS OF PROJECT

The SIGHT project team remained stable and fully staffed throughout FY26. The team is organized into two sub-teams: Advanced Manufacturing (AM) Hub and Artificial Intelligence for Occupational Safety. Leadership includes: (a) Principal Investigator Dr. Fadel M. Megahed; (b) Co-Principal Investigators Drs. Mohammad Mayyas, Arthur Carvalho, and Jay Shan; and (c) Project Manager Dr. Mohamed Farrag. Research staff includes: Senior Personnel Reza Abrishambaf (departed on May 15, 2026, due to increased teaching responsibilities), Associate Research Scientist Dr. Ibrahim Yousif, and GRA Michael Wise. Four hourly student assistants have supported the project so far. The team also receives support from safety consultant Dr. Lora A. Cavuoto and external evaluator, the Discovery Center. Five industry partners remain engaged: MaxByte, MeetKai, Engineered Profiles, Industrious Group Inc., and Yamaha Motor Manufacturing Corporation (Yamaha, a.k.a YMMC).

- **Staffing Updates.** The project completed its planned hiring early and maintained continuity throughout the year. Dr. Yousif joined on August 15, 2025. Mr. Wise, Ms. White, Mr. Hamilton, and Mr. Singh joined on August 25, 2025. Mr. Lyublinski joined on December 11, 2025, as the fourth and final hourly student assistant. Minor onboarding variations were anticipated in the work breakdown structure. They had no impact on deliverables, and the project remained ahead of schedule. The second GRA position was intentionally deferred to FY27 to align expenditures with the sponsor's even-split directive. During Q4, Dr. Reza Abrishambaf concluded his effort on May 15, 2026. His tasks were absorbed by the team without impact on progress.
- **Looking to Year 2.** The team formalized several personnel transitions through the approved Q4 budget revision, effective July 1, 2026: Dr. Yousif transitioned from Postdoctoral Researcher to Associate Research Scientist to retain a high-performing team member; Mr. Hamilton transitioned from an hourly role to a GRA as he begins full-time MS study in Summer 2026; and Mr. Singh and Ms. White completed their roles upon graduation and will be backfilled by new hourly student assistants. Figure XX summarizes this FY26-to-FY27 personnel transition and the Year-2 sub-team structure.

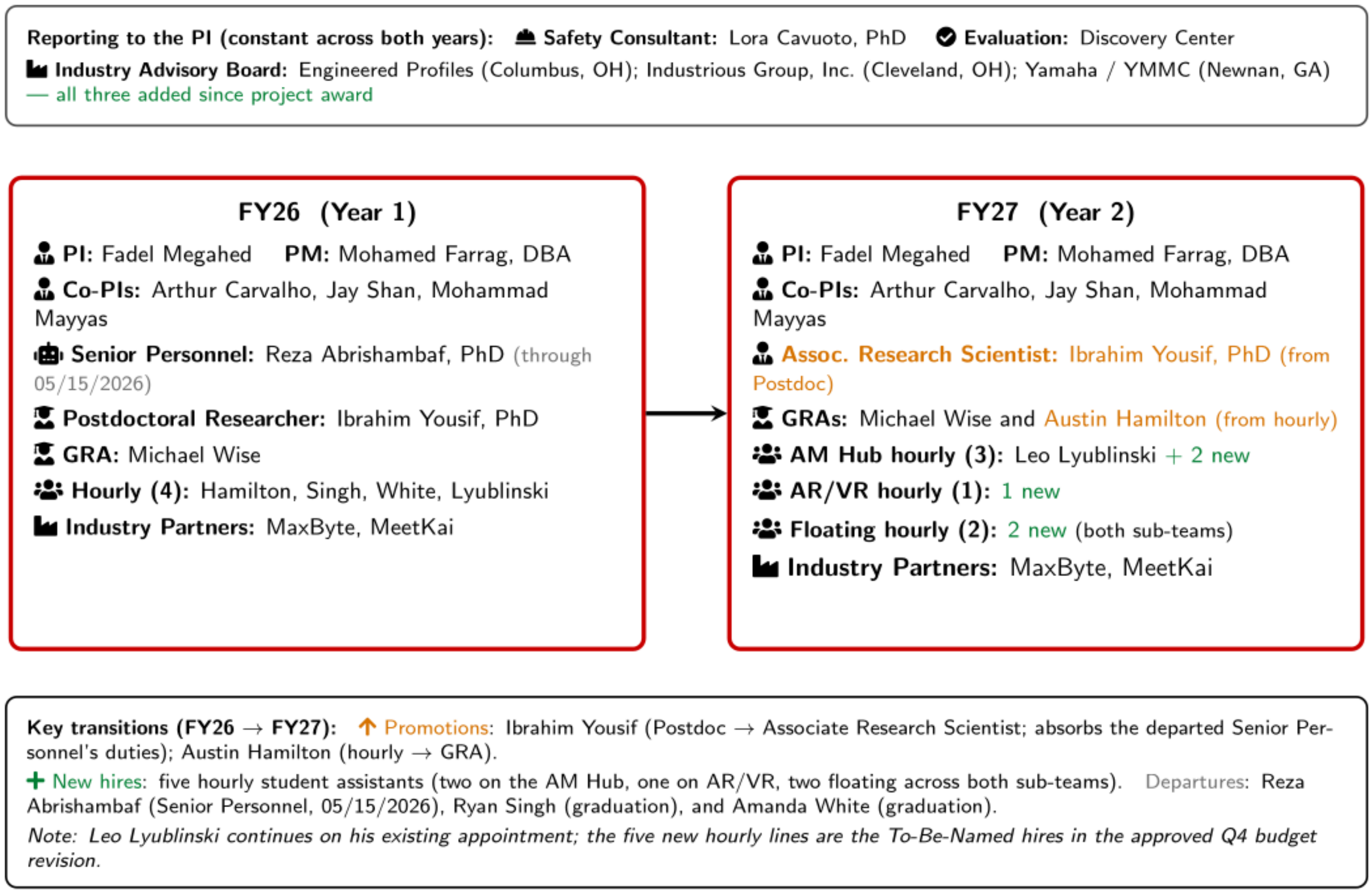


Figure XX. FY27 (Year 2) SIGHT project personnel structure and the FY26 to FY27 transition, organized into the AM Hub and AI/AR/VR sub-teams. Promotions are highlighted in orange, and new hourly hires in green.

- **Non-Financial Administrative Challenges.** No material non-financial administrative challenges were encountered during the reporting period. The project experienced no non-compliance or resource-constraint issues. The project's IRB protocol was approved on November 20, 2025, and remains in good standing. A planned IRB amendment to add student researchers is being consolidated into a single FY27 submission once the Year-2 staffing plan is finalized.

9. PROJECT BUDGET STATUS: ON APPROVED BUDGET REVISION

During the first year (July 1, 2025 – June 30, 2026), the project remained under budget and on or ahead of schedule. **As of Q4, the project is >64.4% complete**. Figure XX presents the FY26 Project Budget Burn Rate, comparing cumulative expenditures to both WSIC-expected and SIGHT-expected trajectories through Q4.

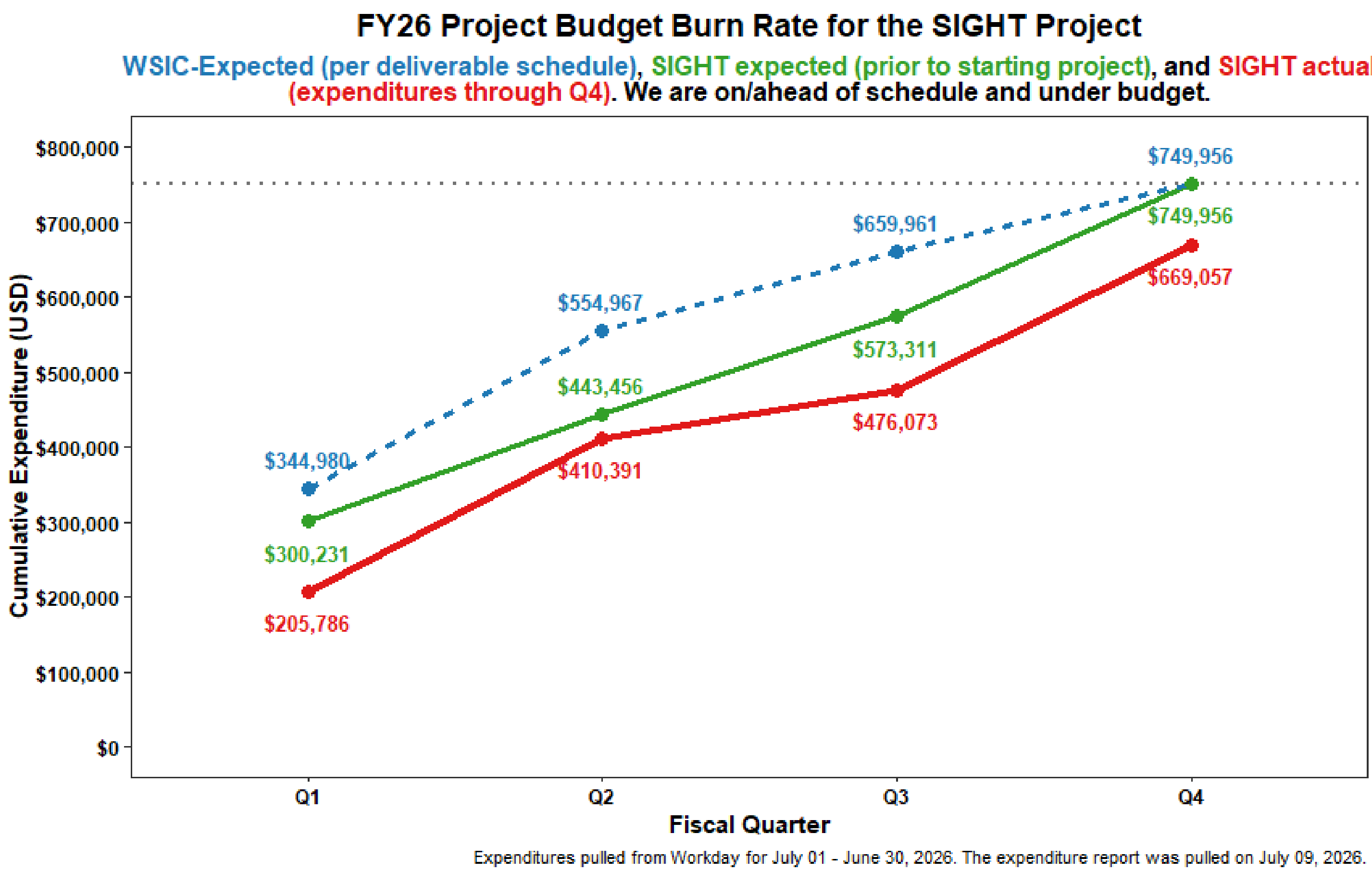


Figure XX. FY26 Project Budget Burn Rate for the SIGHT Project. Cumulative actual expenditures are shown against the WSIC-expected schedule and the SIGHT pre-project projection through Q4. Expenditure data for July 1, 2025 – June 30, 2026 was extracted from Workday on July 9, 2026.

According to the current Workday data, FY26 cumulative spending is **approximately $669,000**, compared to an allocation of **$749,956**. This results in an **underspend of ~$80,000** (~11%) for the first year. These figures are preliminary and will be finalized following Paula Murray's (MU Staff Accountant II) expenditure report review, Q4 invoicing, and close-out on July 31, 2026. The BWC/WSIC directive requires expenditures to be evenly split between FY26 and FY27. This underspend reflects intentional, compliant pacing. Since the BWC reimburses only actual expenditures, the unspent funds remain with the sponsor as cost savings, with **no impact on project scope, schedule, or deliverables**.

- **Fiscal Challenges.** There are no adverse fiscal challenges to report. The FY26 underspend is due to several factors, all previously documented in the Q1–Q3 reports and fully absorbed without affecting technical progress. The contributing factors are:
  - The Postdoctoral Researcher was onboarded approximately 1.5 months later than planned. This delay was fully mitigated through strong team performance.
  - The second Graduate Research Assistant position was intentionally deferred to FY27 to align with the even-split expenditure directive.
    - Minor contract ratification timing in Q1 shifted some partner billing.

    - University policy requiring hourly student workers to be on campus prevented hourly student assistants from working during most of the winter break (January 1–25, 2026). Their responsibilities were reassigned to available faculty researchers, the Postdoctoral Researcher, and the GRA to maintain momentum.
    - The SIGHT team completed several tasks in-house, originally scoped for partner execution, using open-source tools. This reduced external costs while maintaining quality and the project timeline.
- **Budget Revisions.** A budget revision was requested and approved during this reporting period through the fully signed Grant Addendum (Ref. BWC260799), with Miami University and BWC signatures completed on June 30 and July 1, 2026. The total approved project cost remains $1,499,912. The revision reallocates funds among categories to support Year-2 continuity: Personnel +$22,744, Contracts −$40,859, Other Direct Costs +$14,400, and Indirect +$3,714 (auto-adjusted). Travel and Equipment are unchanged. The contract reduction reflects a de-scope of the MaxByte Technologies contract for tasks completed in-house by the SIGHT team using open-source tools. The reallocated funds support Year-2 staffing continuity, including transitioning the Postdoctoral Researcher to an Associate Research Scientist role, promoting an hourly student researcher to Graduate Research Assistant, and adding hourly student support. It also funds AI development tooling subscriptions (i.e., Claude Code to accelerate the software development made by the MU team).
- **Reimbursement Justification.** N/A. Quarterly expenditures and invoices have not exceeded the projected deliverable percentage for reimbursement. Invoice amounts align with progress to date. Therefore, no justification for over-expenditure is required.
- **Equipment and Contracts.** No equipment has been requested or purchased for the SIGHT Project. Both contractors have fully executed, active contracts and remain within budget. After the approved revision, the MaxByte Technologies scope of work has been reduced to $157,255 (consistent with the aforementioned transition to several open-source artifacts), while the MeetKai contract remains unchanged. All previously executed contractor work remains in scope. Per contract, neither contractor will exceed its total budget.
- **Future Budget Considerations.** The approved Q4 revision addresses Year-2 staffing continuity. No further budget revisions or redirections are anticipated at this time. This information is provided for planning purposes only and does not constitute a formal budget revision or redirection request.

10. ACCESSIBILITY CONSIDERATIONS (*TRACK 3 – PROTOTYPE REQUIREMENT ONLY*)

Not applicable.

11. COLLABORATION EFFORTS

[List the industry partners and collaborators involved, specifying their roles. Include stakeholders from various sectors such as industry, academia, government, and the community. Describe any meetings that have occurred during the first half of the project, along with data collection and analysis, project decisions/results, and significant progress made through collaboration.]

12. COMMERCIALIZATION EFFORTS

[Describe the project's current commercialization concept, approach, and efforts towards developing a commercialization strategy (Track 2 – Proof of Concept) or commercialization plan (Track 3 - Prototype). Describe anticipated timelines and discuss the distribution of roles amongst collaborators.

Track 3 – Prototype recipients: *Describe the pathway forward to ensure that the Personal Protective Equipment (PPE) or Personal Protective Technology (PPT) is market ready upon completion of the project.]*

13. PROJECT COMMUNICATIONS, OUTREACH, AND DISSEMINATION

Our team leveraged several communication channels to promote awareness of our SIGHT project and share our progress with both internal and external audiences.

a. **Internal and external stakeholders**: We engaged and shared progress with internal and external stakeholders through weekly sub-team meetings, biweekly full-team meetings including industry partners (MeetKai, MaxByte, Engineered Profiles, Yamaha, and the Discovery Center), and quarterly progress presentations to BWC/WSIC.
b. **Outreach activities**: Project outreach included presentations and demonstrations of our developed systems at the Ohio Safety Congress & Expo (OSC) 2026, where the team engaged representatives from more than 20 organizations, including NASA, the City of Cleveland, and Scotts, generating new opportunities for collaboration and follow-up discussions. Additional outreach included industry site visits (e.g., Engineered Profiles and Yamaha Motor Manufacturing Corporation of America), partner demonstrations, and meetings with academic leaders (e.g., with the Farmer School of Business Advisory Council), highlighting progress on the immersive safety training platforms and AI-powered manufacturing safety tools we are developing.
c. **Public communication**: The project continued to expand its public visibility through the SIGHT website, university communications (e.g., via official press releases and LinkedIn posts), and media coverage (e.g., by Government Technology and Halldale).
d. **Academic dissemination:** Scholarly work has progressed significantly. Our first academic paper related to the AI components of the project was presented at NAMRC 54 and published in Manufacturing Letters, covering our novel AI-powered manufacturing safety chatbot. A follow-up paper was submitted to the HICSS conference and is currently under review. The project team also continues to evaluate additional publication opportunities in leading journals focused on safety, manufacturing, AI, and information systems. We currently have a working paper on novel architectures to validate digital twins, with a particular focus on applications to human-robot collaboration safety.

14. UPDATED COMMUNICATIONS MATRIX

PROJECT TEAM COMMUNICATIONS MATRIX							
Communication Type	**Intended Recipients**	**Description/ Purpose**	**Date Occurred/ Frequency**	**Owner**	**Distribution Method**	**Internal/ External**	**Estimated Reach**
Project Kickoff Meeting	BWC/WSIC, Miami University project team, industry partners, safety consultant, Discovery Center evaluators	Established project vision, team structure, communication protocols, roles, and initial action items	July 1, 2025; once	PI Dr. Fadel Megahed; PM Dr. Mohamed Farrag	Meeting, slides, emails, Google Drive and Notion documentation	Internal and External	Core project team, sponsor, and partner representatives
Weekly Sub-Team Meetings and Biweekly Full-Team Meetings	Miami research team, student researchers, PI, Co-PIs, PM, industry partners, safety consultant, Discovery Center	Coordinated project tasks, reviewed technical progress, assigned next steps, identified risks, and aligned work across AI, VR/AR, advanced manufacturing, evaluation, and industry-partner workstreams	Weekly sub-team meetings and biweekly full-team meetings throughout July 2025–June 2026	PI, PM, Co-PIs, sub-team leads	Zoom/in-person meetings, email, Google Drive, Notion task tracking	Internal and External	Approximately 15–25 recurring project stakeholders
Quarterly BWC/WSIC	BWC/WSIC program managers and fellow awardees	Provided sponsor-facing updates on project progress, deliverables,	Quarterly; Q1–Q4 during	PI, PM, Co-PIs	Quarterly report documents, slide decks,	External	BWC/WSIC staff and awardee audience

Progress Reports and Presentations		risks, technical progress, partner engagement, and upcoming milestones	first project year		email submissions, and sponsor meetings		
Project Communication Plan / Communication Governance	BWC/WSIC, Miami project team, industry partners, evaluators, communication partners	Formalized communication channels, internal/external stakeholder groups, distribution methods, media strategy, social media strategy, website/blog plan, publication strategy, and communication responsibilities	Version 1.0 (Oct. 24, 2025); Version 1.1 (Nov. 5, 2025)	PI and SIGHT leadership team	Project Communication Plan submitted and stored on Google Drive	Internal and External	All core project stakeholders and the sponsor team
Miami University News Release	General public, industry stakeholders, academic audiences, Miami University community	Announced the $1.5M WSIC grant and introduced the SIGHT project's goal of using AI, VR, and AR to reduce preventable manufacturing injuries.	July 18, 2025; once	Miami University Communications and Marketing; PI/project team	Miami University News online article (link)	External	Public online audience
SIGHT Project Website and Blog	General public, BWC/WSIC, academic audiences, industry	Served as the central public platform for project goals,	Website active during the first year;	PI Dr. Fadel Megahed; Co-PIs Dr.	Public website and blog (link)	External	Public online audience

	partners, prospective collaborators, students	announcements, progress updates, and public-facing dissemination. Blog posts included the project funding announcement, summer workshop, quarterly progress, and AI symposium update	Five blog posts published on Apr. 22, 2025; Jul. 21, 2025; Aug. 8, 2025; Dec. 31, 2025; Apr. 17, 2026	Arthur Carvalho and Dr. Jay Shan			
Miami University AI Symposium Presentation	Miami University academic community, AI researchers, faculty, students, and administrators	Presented "SIGHT: AI-Driven Immersive Safety Training for Industry 5.0 Manufacturing," highlighting AI/XR safety training research	Apr. 17, 2026	Co-PI Dr. Arthur Carvalho	University symposium presentation	Internal and External	Approximately 25 attendees
Social Media / LinkedIn Posts	Academic, industry, professional, and public audiences	Shared project announcements and progress updates through the Farmer School of Business and individual project team member LinkedIn accounts	Began July 2025; ongoing	FSB Communications and Marketing; PI; Co-PIs; project team	LinkedIn (links: #1, #2, #3, #4, #5)	External	LinkedIn professional network

University Leadership Amplification	Miami University faculty, staff, senior leaders, industry constituents, and public stakeholders	Project highlighted during President Crawford's State of the University Address and subsequent university communications	Oct. 15, 2025	Miami University President's Office	University address (link) and email/newsletter	Internal and External	University-wide audience
Independent Media Coverage	Public, manufacturing, simulation/training, higher education, and safety communities	The project was featured by independent media following the Miami University announcement, increasing public visibility	Aug. 2025	Government Technology; Halldale Media	Online news articles (links: #1, #2)	External	Public online readership
Engineered Profiles Plant Tour	Miami team, MaxByte, Engineered Profiles	Observed manufacturing operations to improve the realism and applicability of SIGHT technologies	Mar. 13, 2026	PI; Co-PIs; Engineered Profiles	Site visit	External	Project team and partner representatives
Yamaha Advisory Visit	Yamaha advisory partner and Miami project team	Discussed project progress and explored future digital twin and manufacturing collaboration opportunities	Mar. 12, 2026	PI; Co-PIs	AM Hub visit and meetings	External	Project team and Yamaha representatives

Ohio Safety Congress & Expo (OSC) 2026	Safety professionals, manufacturers, public agencies, BWC staff	Demonstrated project progress, gathered stakeholder feedback, and engaged with external organizations	Mar. 2026	PI; Co-PIs; SIGHT team	Conference presentation and demonstrations	External	More than 20 organizations engaged
NAMRC 54 Paper: Manufacturing Safety Chatbot	Academic manufacturing researchers, AI researchers, and industry	Presented research on the multimodal manufacturing safety chatbot and RAG evaluation framework	Jun. 2026	Project Team	Conference presentation and proceedings (link)	External	NAMRC research community
Ohio Safety Congress & Expo (OSC) 2027	Safety professionals, manufacturers, public agencies	Demonstrate the completed VR/AR safety training prototype	Planned 2027	PI; Co-PIs;	Conference presentation and demonstrations	External	Conference attendees
HICSS Conference	AI and information systems researchers	Disseminate submitted research through conference presentation and public preprint, subject to acceptance.	Pending	Project Team	Conference presentation	External	Academic research community
Journal of Intelligent Manufacturing Submission	Manufacturing research community	Submit revised digital twin manuscript describing SIGHT's digital twin framework and validation.	Planned 2026	Project authors	Journal submission	External	Manufacturing research community

15. CHALLENGES

[Provide an updated overview of key challenge areas identified in quarterly progress reports to date. Highlight any trends in recurring issues and describe how they are being addressed. Include any adaptive strategies, escalations, or changes in project approach in response to these challenges.]

16. ACTION PLAN

Nothing to Report.

17. RISK MANAGEMENT LOG

Risk Description	Risk Category	Occurrence	Impact to the Project	Corrective Actions Taken
[Example: 1. IRB approval delays]	*[External]*	*[August/September 2025]*	*[Moderate]*	*[Consulted with more experienced subject matter experts to ensure actions were correct and submitted appropriate supporting documents.]*
<add more rows as needed>				

*[Include a current Risk Management Log documenting risks encountered to date, along with any **newly identified risks** based on activities and actions that have occurred during the first half of the project. Include specific actions and contingency strategies taken to mitigate new risks, and ensure consistency with previously reported risks.]*

18. LESSONS LEARNED

[Describe any lessons learned, by the team or through industry partner collaborations, during the first half of the project. Include any relevant successes or setbacks.]

19. ASSISTANCE REQUEST

None

20. LOOKING AHEAD

In the second year (July 2026 – June 2027) of the SIGHT project, our team will shift focus from foundational AI development to the integration, testing, and fine-tuning of our immersive XR platforms. This phase will emphasize real-world deployment at the Miami University Advanced Manufacturing (AM) Hub and rigorous evaluation with our industry partners.

A. Technical Activities

- **XR Proof-of-Concept Completion**: Building on the Year 1 deployment of the VR environment, we will complete the development of the AR-based real-time hazard coaching system (Task 4, currently in progress). This includes finalized hardware integration for the Magic Leap 2 headsets and the deployment of cohesive, step-by-step Standard Operating Procedure (SOP) use cases. The AR system will prioritize high-impact safety interventions—such as novice CNC lathe operation—that integrate real-time IIoT data and on-headset machine detection.
- **Pilot Testing & Evaluation**: A primary focus of Year 2 will be human-participant testing (Task 5.1). We will recruit participants to evaluate the usability and effectiveness of the VR and AR modules in improving hazard recognition and retention.
- **Deployment & Commercialization Roadmap**: As we transition into Year 2, the project remains on schedule to deliver a scalable roadmap for commercializing our XR innovations. A cornerstone of this strategy is the formalization of our intellectual property, highlighted by the submission of two technology disclosures: the image-to-3D pipeline (submitted April 10, 2026) and the cognitive autonomous robotic assembly system, which is currently moving toward a provisional patent. Our industry partners, MaxByte and MeetKai, continue to play a pivotal role in this phase; MaxByte is leading the commercialization efforts for our image-to-3D invention and exploring real-world applications for the autonomous pallet truck, while MeetKai has successfully transitioned all digital assets to a Miami University repository to ensure platform stability and long-term hosting. Looking ahead, we will leverage feedback from over 20 organizations engaged at the 2026 Ohio Safety Congress and insights from our industry advisors at Yamaha and Engineered Profiles to refine these systems for broader market readiness by June 2027.

B. Administrative Deliverables

- **Quarterly Reporting**: We will submit the subsequent quarterly reports (e.g., Q5, Q6, and Q7) scheduled for October 9, 2026, January 15, 2027, and April 9, 2027.

- **Budget & Staffing Management**: We will implement the Year 2 budget revision, which includes transitioning Dr. Ibrahim Yousif to Associate Research Scientist and Austin Hamilton to a Graduate Research Assistant (GRA) role. We also plan to hire five additional student workers to support AI updates and AM Hub operations.
- **Procurement**: Planned purchases for Year 2 include Claude Code subscriptions, a Roboflow annual subscription, and additional camera hardware for the safety monitoring setup.

C. Planned Team Meetings

- **Bi-Monthly Progress Reviews**: To maintain project momentum, bi-monthly meetings will continue every other Friday. These meetings will continue to serve as checkpoints for technical and partnership updates.
- **Interim Review Session**: A significant portion of the July 10, 2026, meeting will be dedicated to reviewing the draft of the BWC Interim Progress Report.

D. Industry Partner and Consultant Activities

- **MaxByte Inc.**: MaxByte will continue supporting our AR training module development. They will also play a key role in disseminating our findings to manufacturing organizations in OH, and will support the next steps of commercializing our image-to-3D invention per the legal guidance that we will receive from Miami's internal and external legal counsels.
- **MeetKai Inc.**: Following the successful asset handoff to Miami University, MeetKai will provide ongoing support for VR hosting and platform stability while refining the user experience based on Year 1 feedback.
- **Safety Consultant (Lora Cavuoto)**: Lora will lead the expert feedback sessions for the VR modules and collaborate on the finalization of the digital twin research paper for *the Journal of Intelligent Manufacturing*.
- **Industry Advisors (Yamaha Motor Manufacturing & Engineered Profiles):** Both companies will continue to provide feedback on the digital twin environment and the AR application, helping validate realism, functional accuracy, and relevance for future pilot testing.

E. Outreach and Dissemination:

- The second AI Paper "RAG vs. Long Context: A Comparative Study of AI Architectures for Manufacturing Safety Training," was submitted to the HICSS conference's "AI-enabled Digital Transformation for SMEs" track, currently under review. The minitrack cooperates with the "Electronic Markets" journal, in which selected papers can be published as extended versions. If the paper is accepted by HICSS and recommended for *Electronic Markets*, we immediately begin working on a substantial extension. If the paper is not accepted, we will target the upcoming Decision Support Systems (DSS) Special Issue on "Extended Reality (XR) for Enhanced Decision Making and Organizational Transformation."
- The digital twin paper titled "A Cognitive-Emulation Architecture for Validated Decision-Making in Digital Twins: Application to Human-Robot Collaboration Safety" is being developed for submission to the Journal of Intelligent Manufacturing.

21. ADDITIONAL INFORMATION

[Provide any relevant updates or insights from the first half of the project not detailed in previous sections (e.g., team achievements, key learnings or supplemental visuals/documents). If no additional content, state "Nothing additional to report".]